Lecture-21
Dimensional Analysis II: Buckingham Pi Theorem

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Learning Objectives

By the end of this lecture, you should be able to

1. Apply the **Buckingham Pi Theorem** to determine the number of independent dimensionless groups required for a problem.
2. **Select appropriate repeating variables** based on structural criteria.
3. Define and interpret the physical significance of **core dimensionless numbers** in fluid mechanics.
4. Differentiate between **geometric, kinematic and dynamic similarity**.

Buckingham Pi Theorem

- The dimensions that are **commonly chosen as a basis** to measure other quantities are called **primary dimensions**.

> There are many possible choices of base physical dimensions. The SI standard selects the following dimensions and corresponding **dimension symbols**:
>
> time (T), length (L), mass (M), electric current (I), absolute temperature (Θ), amount of substance (N) and luminous intensity (J).

- The Buckingham-Pi theorem states that we can **transform a relationship between n-parameters** into a corresponding relationship **between n-m independent and dimensionless parameters**, where m is the number of independent primary dimensions.

https://en.wikipedia.org/wiki/Dimensional_analysis

Geometric, Kinematic and Dynamic similarity

- Geometric similarity means the **model and the prototype have the same shape** and are proportional in all linear dimensions.
- Kinematic similarity means the **motions of the fluid or system are proportional** at corresponding (homologous) points.
- Dynamic similarity means the **ratios of all forces acting on corresponding particles or surfaces are identical** in both the model and the prototype.
- Dynamic similarity is practically achieved by **making dimensionless parameters**—like the Reynolds Number or Froude numbers **equal in both systems**.

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Revisiting the learning objectives

1. Apply the **Buckingham Pi Theorem** to determine the number of independent dimensionless groups required for a problem.
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4. Differentiate between **geometric, kinematic and dynamic similarity**.

